

Capstone Design Manager

Continuing and hardening the UA senior-design project platform

Purpose:

A web app where sponsors submit projects, students rank their choices, and the instructor assigns teams, all in one place instead of email and spreadsheets.

Team

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Noland Miller**

Sponsor

**Dr. Monica Anderson Herzog, Computer Science, The
University of Alabama**

The problem and who it serves

Problem / Need

- Each semester the capstone instructor collects sponsor project proposals, gathers student preferences, and builds team assignments by hand across email, forms, and spreadsheets
- The process is slow, error-prone, and hard to hand off; there is no single place where sponsors, students, and the instructor can see current status.
- The department needs a maintainable platform that a new team can run each year

Users

Students

Browse projects, rank top 3, see assignment

Sponsors

Submit and edit projects, share files, give feedback

Admin (instructor)

Approve, match, email

Stakeholders

- Dr. Monica Anderson (sponsor and primary user)
- CS and Cybersecurity capstone students
- Industry and campus sponsors
- Future capstone teams inheriting the code

Proposed solution and initial scope

Approach

Build a new web platform from the ground up, designed around the sponsor's actual semester workflow: submit, review, rank, assign, communicate. A prior team's repository exists and will be used as a reference for requirements and lessons learned, not as a codebase we extend.

Initial scope

1. Sponsor project submission with review and approval by the instructor
2. Student project browsing and ranked preference submission
3. Assignment tool that matches students to projects from their rankings
4. Role-based access for students, sponsors, and admin
5. Email notifications for key steps (proposals open, preferences due, assignments posted).

Not in scope for now: sponsor feedback and progress tracking, support for other engineering departments, UA single sign-on. Priorities to be confirmed with the sponsor.

Product Backlog & Iteration 1 Planning

Major Product Backlog Themes

-  **Authentication & Role Security** — harden Auth0 signup flow
-  **Project Lifecycle** — proposals, approval, semester rollover
-  **Student Matching** — preference ranking & assignment
-  **Sponsor Engagement** — outreach, updates, feedback
-  **Admin Dashboard & UX** — tables, presentation view
-  **Quality & DevOps** — tests, Docker, CI/CD

Highest-Priority Issues

- 1. Fix Auth0 role-assignment vulnerability**
Privilege escalation risk in student/sponsor signup
- 2. Resend verification email for sponsors**
Resolve issue where verification links expire in 7 days
- 3. Clear error for students not in class roster**
Prevent silent failures during student login flow

 Backlog (initial)	 Iteration 1 (Sep 1–14)
MySSO student login	 Fix role-assignment flaw
Semester rollover & archiving	 Resend verification email
Sponsor feedback improvements	 Roster error message
Dashboard & table polish	 Stand-Up #1 — Sep 1 
CI/CD & test coverage	
Snapshot as of Sep 3 — Iteration 1 committed to security-critical signup fixes. Team: Jayden White, Noland Miller, Margulan Baizhakyp, Gaurav Shrivastava	

Architecture & Technology Approach

Decided

- Web application built from the ground up
- Role-based access for students, sponsors, and admin
- Auth0 for user authentication
- Docker-based development environment
- Prior project used as a reference, not the codebase

Architecture Goal

Keep the system maintainable, secure, and easy for future capstone teams to continue developing.

Still Investigating

- Frontend/backend framework and API structure
- Database schema for projects, semesters, and rankings
- CI/CD and production deployment
- Feasibility of future UA MySSO integration

Quality Strategy

Testing

- **Baseline tests for core user workflows**
- **Test new functionality as features are added**
- **Verify major fixes before merging**

Development Workflow

- **Separate feature/fix branches**
- **Pull requests before merging**
- **Team review for significant changes**
- **Issues linked to development work**

Definition of Done

- **Acceptance criteria met**
- **Functionality tested**
- **PR reviewed and merged**
- **No known blocking issues**

Documentation Expectations

README, GitHub Issues, Project views, and technical documentation updated as the project changes.

Risks & Dependencies

Risks

- **Bad Auth0 integration and authorization design → Unauthorized access to data**
- **Django/Vue API inconsistencies → Disconnect between backend/frontend**
- **Docker inconsistencies → Difficult deployments and debugging**

Dependencies

- **Authorization (Auth0, Django, Vue)**
- **Database access (Django, Vue)**
- **Deployments (Google Cloud)**

Links

- [Github](#)